Prediction of Final Wildfire Burned Area on Detection in New Zealand Using Machine Learning

Project Proposal

by

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Abstract

Wildfires in Aotearoa New Zealand can end up being very different in size, even when the conditions at the time of detection appear similar. Current fire-danger measures, such as the Fire Weather Index (FWI), are useful for describing how favourable the conditions are for fire activity, but they do not directly estimate how large an individual fire will eventually become. This research looks at whether machine learning can be used to predict the final burned area of a wildfire from information available at or around the time it is detected. The study uses the New Zealand Wildfire Occurrence Database (WOD), which contains wildfire records together with weather, fuel, terrain, location and time-related information. Final burned area, recorded in the `AreaHa` variable, is treated as a continuous regression target. Several regression models will be trained and compared using a chronological split, where earlier fires are used for training and later fires are kept for testing. This is intended to give a more realistic indication of how the models could perform when applied to future fires. A major challenge is that most fires in the dataset have very small burned areas, while only a small number grow into large fires. The study will therefore examine target transformation and imbalanced-regression methods to see whether they can improve predictions for these rare large fires. Model performance will be measured using MAE, RMSE and R^2, together with separate analysis of larger fires. Feature importance and SHAP analysis will be used to identify the variables that have the greatest influence on the predictions. The study aims to determine how useful machine learning is for predicting final wildfire burned area in New Zealand and to identify the main challenges involved in predicting rare large fires.

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Chapter 1

Introduction

Wildfires can vary significantly in size, even when they occur under similar environmental conditions. Some fires are brought under control while they are still small, whereas others continue to grow and become much larger events. Understanding why some fires grow much more than others is important for wildfire planning and management. In Aotearoa New Zealand, fire danger has traditionally been assessed using weather-based measures such as the Fire Weather Index (FWI), which provides an indication of how favourable the conditions are for fire development [Renwick et al., 2007, Pearce et al., 2007]. However, a high level of fire danger does not necessarily indicate how large a particular wildfire will eventually become. This research investigates whether machine learning can be used to predict the final burned area of a wildfire using information available at or around the time it is detected. This chapter introduces the research problem, explains its significance, and presents the aim and objectives of the research.

1.1 Problem Statement

Predicting how large a wildfire will become is difficult because fire behaviour is influenced by several factors at the same time. Weather conditions such as temperature, relative humidity, wind and rainfall can affect fire development. However, factors such as available fuel, topography, location and other environmental conditions can also influence how a fire develops. Because these factors

can interact with each other, there is not always a straightforward relationship between an individual environmental variable and the final size of a fire.

In New Zealand, the Fire Weather Index system provides a useful measure of environmental fire danger and has been used to support fire management decisions [Renwick et al., 2007]. However, the FWI mainly describes how favourable the conditions are for fire activity. It does not directly indicate the final burned area of a particular fire. For example, a fire detected during a period of high fire danger may still remain small because there is limited fuel available, the location is less favourable for fire growth, or the fire is suppressed quickly. On the other hand, a fire that starts under less extreme weather conditions may still grow considerably when other environmental factors are favourable.

Machine learning has recently been used to study different wildfire-related problems, including wildfire occurrence, susceptibility, fire danger, fire spread and final fire size. For example, Coffield et al. [Coffield et al., 2019], investigated the prediction of final fire size at the time of ignition. More recent studies have also looked at burned-area prediction and the challenge of predicting rare large fires [Pang et al., 2025, Camacho et al., 2026]. However, many of these studies use data from regions outside New Zealand, including Alaska, Siberia and the western United States. The relationships found in these regions may not necessarily apply to the environmental and geographical conditions of New Zealand.

The New Zealand Wildfire Occurrence Database (WOD) provides an opportunity to study this problem using data specific to New Zealand. The WOD combines wildfire records with information about weather, fuel, topography and geographical conditions [Kiely et al., 2026]. The version of the database used in this research contains wildfire records through 2025, providing a large collection of historical observations that can be used to develop and evaluate prediction models.

Another challenge comes from the distribution of wildfire burned area in the dataset. Most fires have very small burned areas, while only a small number of fires become extremely large. This makes the problem difficult because a model could achieve a relatively good overall error by performing well on the many small fires while performing poorly on the few large fires. Since these large fires are often the cases of greatest interest for wildfire management, it is important to consider them separately. Recent research on imbalanced regression and

extreme wildfire prediction has also highlighted the importance of addressing these rare cases [Camacho et al., 2022, Camacho et al., 2026, Koh, 2023].

Therefore, this research focuses on predicting the continuous final burned area of an already detected wildfire using environmental, geographical and temporal information available at or around the time of detection. The research also considers the additional challenge of predicting the relatively small number of large wildfires.

1.2 Problem Significance

Predicting the final burned area of a wildfire is important because the impact of a fire depends not only on whether it occurs, but also on how large it becomes. A small fire and a fire that grows to hundreds or thousands of hectares can require very different levels of firefighting resources. They can also have very different effects on communities, ecosystems and infrastructure.

Current fire-danger measures provide useful information about the environmental conditions in which a fire occurs, but they do not directly estimate how large an individual wildfire will eventually become. A model that predicts burned area could provide additional information once a fire has been detected. For example, if a newly detected fire is predicted to have a higher potential for becoming a large event, this information could help support decisions about the level of attention and resources that may be needed. Such a model would not replace the judgement of fire-management teams, but it could provide another source of information when assessing the potential impact of a fire.

The problem is also interesting from a machine learning perspective. The wildfire dataset contains different types of information, including weather, fuel, terrain, spatial and temporal variables. These factors may interact in ways that are difficult to capture using a simple linear relationship. Previous wildfire studies have shown that machine learning methods can be useful for identifying complex relationships between environmental conditions and wildfire outcomes [Jain et al., 2020, Malashin et al., 2025]. However, the large difference between the number of small and large fires creates an additional challenge. Large fires are uncommon in the dataset, but they are often the cases that matter most from a wildfire-management perspective.

This research therefore looks at two related issues. The first is whether machine learning can effectively predict the final burned area of wildfires using New Zealand-specific data. The second is whether the modelling approach can better account for the small number of large fires instead of mainly optimising predictions for the much larger number of small fires. Together, these aspects can help assess how useful machine learning may be for wildfire burned-area prediction in the New Zealand context.

The results of this research may be useful for researchers studying wildfire prediction as well as organisations involved in wildfire planning and management. The findings may also provide a starting point for future work that integrates machine learning predictions into operational fire-management systems.

1.3 Project Aim and Research Objectives

Project Aim: This project aims to develop and evaluate machine learning models for predicting the final burned area of a wildfire in Aotearoa New Zealand using information available at or around the time the fire is detected, with particular attention to the prediction of rare large fires.

The specific research objectives (ROs) are:

- **RO1:** Prepare and analyse the New Zealand Wildfire Occurrence Database to identify suitable environmental, geographical and temporal variables for predicting wildfire burned area.
- **RO2:** Develop and compare different regression models for predicting the continuous final burned area of wildfires using a chronological training and testing approach.
- **RO3:** Investigate approaches for handling the highly imbalanced distribution of wildfire burned area and evaluate their effect on predictions of rare large fires.
- **RO4:** Evaluate the performance of the developed models using appropriate regression metrics and separate analysis of larger wildfire cases.

- **RO5:** Analyse the most important predictors used by the final models to improve understanding of the environmental and geographical factors associated with larger wildfire outcomes.

RO1 focuses on understanding and preparing the WOD before developing the prediction models. The dataset will be examined for missing values, unusual values, changes in feature availability over time and the distribution of burned area. Predictor variables will then be selected based on their relevance to the research problem, whether they would have been available at the prediction time, and the possibility of information leakage.

RO2 focuses on developing and comparing the prediction models. In this study, burned area will be treated as a continuous regression target rather than being divided into predefined fire-size categories. Several regression approaches, including linear and tree-based ensemble models, will be compared. The data will be split chronologically, with earlier wildfire records used for training and later records used for testing.

RO3 addresses the strong imbalance in the wildfire data. Most records represent very small fires, while large fires are relatively uncommon. Different approaches will be investigated to determine whether the models can improve their predictions for these rare large fires without unnecessarily reducing their performance on smaller fires. This will include investigating target transformation and imbalanced-regression approaches.

RO4 will evaluate the models using metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2. Overall model performance will be considered, but the predictions will also be examined across different burned-area ranges. This will help determine whether the models perform reasonably well on larger fires rather than allowing their performance to be hidden by the much larger number of small fires.

RO5 focuses on understanding the predictions made by the models. Feature importance and, where appropriate, SHAP analysis will be used to investigate which weather, fuel, terrain, spatial and temporal variables contribute most to the predictions. This analysis will help determine whether the models are capturing meaningful patterns in the wildfire data and provide some insight into the factors associated with larger wildfire outcomes.

Together, these objectives cover the main stages of the research, starting with understanding and preparing the data, followed by model development and evaluation, and finally interpreting the factors that influence the predictions. The next chapter describes the methodology used to address these objectives, including data preprocessing, feature selection, target transformation, temporal validation, model development and evaluation.

Chapter 2

Literature Review

This chapter reviews previous research on wildfire burned-area prediction and the use of machine learning in wildfire research. It begins with some background on wildfire prediction in Aotearoa New Zealand and then introduces the main types of machine learning approaches relevant to this research. The chapter then reviews previous studies covering wildfire occurrence, fire danger, fire size and burned-area prediction. Finally, the studies are compared to identify the main limitations and research gaps that motivate the present study.

2.1 Background

2.1.1 Wildfire Burned-Area Prediction

The size of a wildfire is influenced by many factors, rather than by a single environmental condition. Weather, available fuel, topography, ignition characteristics and the surrounding environment can all affect how a fire develops. Because these factors can interact with each other, similar weather conditions do not always result in fires of a similar size.

In New Zealand, fire weather has traditionally been monitored using the Fire Weather Index (FWI) system. The system uses variables such as temperature, relative humidity, wind speed and rainfall to describe conditions that are favourable for fire ignition and spread. The FWI forms an important part of the New Zealand Fire Danger Rating System and is used to support fire man-

agement decisions [Renwick et al., 2007]. Earlier research in New Zealand also found that combinations of high temperature, strong winds, low humidity and dry conditions can lead to periods of high fire danger [Pearce et al., 2007].

New Zealand also has some characteristics that make its wildfire environment different from many of the regions commonly studied in the international literature. Pearce et al. reported considerable regional differences in fire danger across New Zealand, while more recent research suggests that climate change may increase the frequency and severity of dangerous fire-weather conditions in some parts of the country [Pearce et al., 2007,Melia et al., 2022]. The Ministry for the Environment has also reported that wildfire risk varies between regions and is influenced by factors such as weather and fuel conditions [Ministry for the Environment, 2025].

These differences are important when considering wildfire size. A fire can start during a period of high fire danger but still remain small because of limited fuel, its location or rapid suppression. Similarly, a fire that starts under less extreme weather conditions can become large when the surrounding fuel and environmental conditions support continued growth.

For this reason, the focus of this research is not simply to determine whether fire danger is high. Instead, it is to predict the final burned area of a fire that has already been detected. The target is therefore a burned area value, measured in hectares, using information that would be available at the time of detection.

2.1.2 Wildfire Data in New Zealand

One of the challenges in developing machine learning models for wildfire research is obtaining enough historical data with consistent information. In New Zealand, wildfire records have historically been collected by different organisations, meaning that information has not always been available in one consistent dataset.

Kiely et al. [Kiely et al., 2026] addressed this issue by developing the New Zealand Wildfire Occurrence Database (WOD). The database brings together wildfire records with spatial, temporal, environmental, topographic, fuel and meteorological information. TThe published WOD contains 114,142 wildfire records from 1998 to 2022, along with 2,613 older fire records dating back to 1931.

The database version used in this study has been updated to include records through 2025, with a total of 125,385 wildfire records. The authors also point out that historical wildfire records can contain uncertainty and errors, which need to be considered when using the data for modelling.

The WOD is particularly useful for this research because it combines several types of information that are normally considered separately. For example, weather conditions can be studied together with fuel, topography and geographical information. This provides an opportunity to investigate whether a combination of these factors can help explain differences in the final size of wildfires.

The version of the WOD used in this research contains records through 2025. This provides a more recent set of observations than the period covered in the published WOD paper. However, the availability of different variables is not consistent across the entire time period. This needs to be considered when selecting features and designing the model evaluation strategy.

2.1.3 Machine Learning

Machine learning is a data-driven approach in which a model learns patterns from existing data and uses those patterns to make predictions for new observations. Unlike a traditional approach where relationships have to be defined explicitly, a machine learning model can learn these relationships from examples in the data.

Machine learning has been used in wildfire research since the 1990s, and its use has increased as larger environmental and remote-sensing datasets have become available. Jain et al. [Jain et al., 2020] reviewed hundreds of studies covering areas such as wildfire detection, occurrence, fire weather, fire behaviour, fire effects and fire management. Their review identified methods such as Random Forest, decision trees, support vector machines and neural networks among the commonly used approaches.

Machine learning methods can generally be grouped into supervised learning, unsupervised learning and reinforcement learning. The appropriate approach depends on whether target values are available and what type of problem is being studied.

2.1.3.1 Supervised Learning

In supervised learning, the model is trained using observations for which both the input variables and the desired output are known. The purpose is to learn a relationship between the input variables and the target so that the model can make predictions for new observations.

Supervised learning is commonly divided into classification and regression. Classification is used when the output belongs to a category, such as predicting whether a fire will occur. Regression is used when the output is continuous, such as predicting rainfall, temperature or, in this research, burned area.

Two supervised learning approaches that are particularly relevant to this research are decision trees and ensemble methods.

A decision tree divides the data using a series of conditions on the input variables. Each split attempts to create groups that are more similar with respect to the target value. Decision trees are relatively easy to understand, but an individual tree can become too closely fitted to its training data.

Random Forest addresses this problem by combining predictions from many decision trees. Breiman [Breiman, 2001] showed that introducing randomness when building the trees and combining their predictions can improve generalisation. Random Forest can also capture non-linear relationships and interactions between variables without requiring a specific mathematical relationship to be defined beforehand.

Another important approach is gradient boosting. Rather than building independent trees and averaging their predictions, boosting builds trees sequentially. Each new tree attempts to improve the errors made by the previous trees. Friedman [Friedman, 2001] introduced a general gradient boosting framework for regression and classification. XGBoost later extended this approach through an efficient implementation designed for large datasets [Chen and Guestrin, 2016].

2.1.3.2 Unsupervised Learning

Unsupervised learning is used when there is no known target variable. Instead, the model attempts to identify patterns or structures within the input data.

Clustering is one common example. A clustering algorithm groups observa-

tions that are similar according to their input features. In a wildfire context, for example, fires could be grouped according to similar environmental conditions without specifying the groups in advance.

K-means is a commonly used clustering algorithm that groups observations according to their distance from cluster centres. Although unsupervised learning can be useful for exploring the wildfire dataset, it is not the main approach for this research because the historical WOD records already contain the final burned area for each fire.

2.1.3.3 Reinforcement Learning

Reinforcement learning differs from supervised and unsupervised learning because an agent learns by interacting with an environment. The agent takes actions and receives rewards or penalties depending on the results of those actions.

Reinforcement learning could have applications in wildfire management, particularly for problems involving decisions about how resources should be allocated over time. However, it is outside the scope of this research because this study focuses on predicting a continuous wildfire outcome rather than learning a sequence of management actions.

2.2 Related Work

Previous machine learning research in wildfire science covers a wide range of problems. These include predicting whether a fire will occur, mapping wildfire susceptibility, forecasting fire danger, predicting fire spread and estimating final burned area. The studies most relevant to this research are discussed below.

2.2.1 Wildfire Weather and Risk Modelling

Traditional wildfire prediction has relied heavily on weather-based indices. In New Zealand, the FWI system has been used for many years to describe fire danger using weather observations. Renwick et al. [Renwick et al., 2007] investigated approaches for predicting fire weather and fire danger in New Zealand, while Pearce and colleagues examined longer-term fire-weather patterns and the effects of climate variability on fire danger [Pearce et al., 2007].

These approaches are useful because they provide relatively simple measures that can be understood and used in operational fire management. However, they mainly describe the conditions surrounding a fire and do not directly predict the final size of an individual wildfire.

More recent research has combined weather information with machine learning. Kondylatos et al. [Kondylatos et al., 2022], for example, used machine learning to predict next-day wildfire danger using meteorological, vegetation and human-related information. Their results suggested that variables such as soil moisture and relative humidity could provide useful information in addition to traditional fire-weather indices.

Ardid et al. [Ardid et al., 2025] used a Random Forest model with high-frequency weather-station observations to forecast fire potential. Their results suggested that sub-hourly weather information could improve predictions of pre-fire conditions compared with a standard fire-danger index. A later study extended this work across several Australian regions and found that the machine learning approach continued to outperform the Fire Behaviour Index while also considering the economic value of improved warnings [Ardid et al., 2026].

Di Giuseppe et al. [Di Giuseppe et al., 2025] considered wildfire activity at a global scale using a data-driven prediction system. An important finding from their work was that the quality of the input data had a larger effect on forecast performance than simply increasing the complexity of the machine learning model.

This finding is relevant to the current study. Initial exploration of the WOD shows that some variables contain substantial missing data and that the availability of certain fire-weather variables changes over time. Therefore, including more variables does not necessarily mean that the resulting model will be better.

2.2.2 Machine Learning for Wildfire Occurrence and Susceptibility

A large part of the wildfire machine learning literature focuses on predicting where or when a wildfire is likely to occur rather than predicting how large an individual fire will eventually become.

Jain et al. [Jain et al., 2020] identified wildfire occurrence, susceptibility and

risk as major application areas for machine learning. These studies commonly use combinations of weather, vegetation, topography, land cover and human-related variables.

Tonini et al. [Tonini et al., 2020], for example, used Random Forest to produce wildfire susceptibility maps for the Liguria region of Italy. Their model used land cover, vegetation, roads, altitude and other environmental variables. The study also considered spatial cross-validation, highlighting the importance of accounting for spatial dependence when evaluating susceptibility models.

Shmuel and Heifetz [Shmuel and Heifetz, 2022] used several machine learning models to predict global wildfire occurrence and also developed regression models for burned-area size. Their results showed that machine learning methods could improve both fire-occurrence and burned-area predictions compared with simpler linear approaches.

Other studies have also compared methods such as Random Forest, support vector machines, neural networks and hybrid approaches for wildfire susceptibility and risk modelling [Jain et al., 2020, Jaafari et al., 2019]. Overall, these studies suggest that non-linear models can be useful when wildfire behaviour depends on interactions between several environmental factors.

However, wildfire susceptibility is different from the problem considered in this research. Susceptibility models generally estimate whether a particular location is likely to experience a fire, whereas this study begins with an already detected wildfire and attempts to predict its final burned area.

This distinction is important because the prediction target and the available information are different. A model that is effective at identifying locations where fires are likely to occur does not necessarily provide an accurate estimate of how large a particular fire will become.

2.2.3 Prediction of Fire Size and Burned Area

Research on final fire size and burned area is more directly related to the current study. Coffield et al. [Coffield et al., 2019] investigated whether machine learning could predict the final size of an Alaskan wildfire at the time of ignition. The study used environmental and fire-related information available at ignition and compared machine learning models for predicting fire-size categories.

The study is important because it shows that wildfire size may be predictable before the final extent of a fire is known. However, the research was conducted in Alaskan boreal forests and treated fire size as categories rather than predicting the exact final area as a continuous value. Therefore, its findings provide useful evidence for the general approach but do not directly address the New Zealand regression problem.

Pang et al. [Pang et al., 2025] developed Fire-Image-DenseNet (FIDN), a deep learning model for predicting final burned area in the western United States. The model combined early fire-spread information with meteorological, geographic and vegetation data. A key strength of this study is that it directly predicts final burned area rather than focusing only on fire occurrence. The study also showed that deep learning could provide accurate predictions while being considerably faster than some physics-based approaches.

However, FIDN uses information from the first three days of fire spread. This is different from the prediction setting in the present study, where the focus is on information available at or around the time a fire is detected. This distinction matters because information about how a fire has already developed can provide additional information about its eventual size.

Malashin et al. [Malashin et al., 2025] investigated wildfire size using machine learning in Siberian forests. Their study used meteorological, forest and geographical variables and compared several machine learning methods. XGBoost produced the strongest results in their classification experiments, while feature importance and SHAP analysis were used to examine the main factors influencing the predictions.

The study demonstrates the value of combining weather, geographical and vegetation information. However, the model was developed for a particular Siberian environment and treated fire size as a classification problem. Its results therefore cannot be assumed to apply directly to New Zealand or to continuous burned-area prediction.

Lampe et al. [Lampe et al., 2026] provide another perspective through BuRNN, a recurrent neural network model for predicting monthly global burned area. The model used climate, vegetation and socio-economic variables and was evaluated against process-based fire models.

BuRNN demonstrates the potential of machine learning for burned-area mod-

elling at a large spatial and temporal scale. However, its prediction target is monthly burned area on a global grid, which is quite different from predicting the final area of an individual wildfire.

These studies show that machine learning can be used to estimate fire size or burned area, but they differ in their prediction targets, geographic regions and the information available when making predictions. As a result, their reported results cannot be directly compared with one another or assumed to apply to New Zealand.

2.2.4 Extreme Fires and Imbalanced Regression

One of the main challenges in this research is the small number of large wildfires. Most records in the dataset represent very small fires, while only a small proportion become extremely large. This creates a problem that is related to class imbalance, but the target variable itself is continuous.

Camacho et al. [Camacho et al., 2022] investigated this issue using Geometric SMOTE for regression. Their approach treats rare values in a continuous target as important parts of the distribution and proposes a resampling method designed to improve predictions for rare and extreme values.

This idea is particularly relevant to wildfire burned-area prediction. A standard regression model may obtain a reasonable overall error by predicting the large number of small fires well, while performing poorly on the relatively few large fires. From a wildfire-management perspective, this may not be desirable because large fires can have much greater consequences.

Camacho et al. [Camacho et al., 2026] later applied imbalanced-regression techniques to wildfire final burned-area prediction. Their framework combined machine learning with approaches designed to improve predictions for rare, high-impact wildfire events. This is one of the studies most closely related to the current research because it considers final burned area at an early stage and explicitly addresses the problem of imbalanced regression.

However, the study was conducted using a different wildfire dataset and environmental setting. Therefore, it remains necessary to investigate whether similar approaches are effective when applied to the New Zealand WOD.

Other approaches have also been used to model extreme wildfire behaviour.

Koh [Koh, 2023] combined gradient boosting with extreme-value theory to predict wildfire counts and sizes across the contiguous United States. Spatial cross-validation was also considered to provide a more realistic estimate of model performance.

D'Arcy et al. [D'Arcy et al., 2023] developed a statistical approach for predicting wildfire extremes by focusing on the extreme part of the wildfire distribution. Cisneros et al. [Cisneros et al., 2023] combined spatial statistical modelling with machine learning to predict extreme wildfire frequencies and sizes and considered issues such as zero-inflated data and missing observations.

Taken together, these studies suggest that rare large fires should not simply be treated as ordinary observations. Target transformations, resampling, specialised loss functions, extreme-value methods and evaluation focused on large events are all possible ways of addressing this issue.

2.2.5 Temporal and Spatial Validation

The way a wildfire model is evaluated can have a significant effect on the reported results. A random train-test split assumes that observations are sufficiently independent. However, wildfire records can be related through both time and location.

Gholami et al. [Gholami et al., 2021] highlighted the importance of handling temporal information carefully in wildfire prediction. They also noted that using predictor information from the same time step as the event can lead to an unrealistic evaluation if the model is intended to make predictions in the future.

Koh [Koh, 2023] similarly found that spatial cross-validation can provide a more realistic estimate of model performance than a naive cross-validation approach in wildfire prediction.

BuRNN also used region-based validation because burned-area observations can have spatial dependence [Lampe et al., 2026]. Tonini et al. [Tonini et al., 2020] likewise considered spatial cross-validation when developing wildfire susceptibility models.

These studies suggest that the validation method should reflect how the model is expected to be used. Since the current research aims to use historical wildfire records to predict later wildfire outcomes, a chronological split is more appro-

priate than simply randomly dividing the data. This is particularly important for the WOD because the availability of some predictor variables changes over time.

2.2.6 Interpretability and Understanding Fire Drivers

Prediction accuracy is important, but it is not the only consideration when machine learning is used for wildfire research. It is also useful to understand which variables are influencing the model's predictions. This can help determine whether the model is learning meaningful relationships from the data.

Malashin et al. [Malashin et al., 2025], for example, used SHAP analysis to investigate how environmental variables contributed to wildfire-size predictions. Other wildfire studies have similarly used feature importance to examine the influence of weather, vegetation and terrain.

SHAP was introduced by Lundberg and Lee [Lundberg and Lee, 2017] as a general framework for explaining individual model predictions. It provides feature contributions that can help explain why a model produces a particular prediction.

Interpretability is particularly relevant to this research because a high-performing model should not simply be treated as a black box. If weather, fuel or topographic variables consistently appear as important predictors, this may provide useful insight into the conditions associated with larger fires. However, feature importance does not necessarily mean that a variable causes the outcome, particularly when several variables are correlated.

2.3 Comparative Analysis

The studies reviewed above differ in several important ways. Some predict whether fires are likely to occur, while others focus on fire danger, susceptibility, fire spread or final burned area. The spatial scale also varies considerably, from individual fires to regional areas and global datasets.

A major difference is between fire-danger prediction and fire-size prediction. Studies such as Kondylatos et al. [Kondylatos et al., 2022] and Ardid et al. [Ardid et al., 2025] aim to identify conditions under which fire activity is more likely.

These approaches are valuable for early warning, but they do not directly estimate the final size of an individual wildfire.

Studies such as Coffield et al. [Coffield et al., 2019], Pang et al. [Pang et al., 2025] and Camacho et al. [Camacho et al., 2026] are more closely related to the current research because they consider the final size of a fire. However, they use different types of information. For example, Pang et al. use information about fire spread during the first three days, whereas Coffield et al. and Camacho et al. focus more closely on information available at ignition or during the early stage of a fire.

There is also a difference in model complexity. Deep learning models can handle complex spatial and temporal information, as demonstrated by FIDN and BuRNN [Pang et al., 2025, Lampe et al., 2026]. However, more complex models also require suitable training data. Di Giuseppe et al. [Di Giuseppe et al., 2025] provide an important reminder that improving the quality of input data can be more useful than simply increasing model complexity.

For the present study, this suggests that a complex deep learning model is not necessarily the best starting point. The WOD is a structured tabular dataset containing weather, fuel, terrain and spatial information. Tree-based ensemble models are therefore a reasonable choice because they can capture non-linear relationships and interactions while working well with structured data.

The treatment of rare large fires is another important difference between studies. While some research focuses mainly on overall predictive performance, studies such as Camacho et al. [Camacho et al., 2022, Camacho et al., 2026], Koh [Koh, 2023], D'Arcy et al. [D'Arcy et al., 2023] and Cisneros et al. [Cisneros et al., 2023] pay greater attention to extreme values. This is particularly relevant to the current study because the majority of the WOD records represent small fires.

Overall, the literature does not point to one single machine learning method that is best for every wildfire prediction problem. The choice depends on the prediction target, available data, prediction time, spatial scale and intended use of the model.

2.4 Research Gaps and Challenges

The literature review identifies several gaps that provide the motivation for this research.

First, much of the wildfire machine learning literature focuses on predicting fire occurrence or susceptibility rather than the final burned area of an individual wildfire. Jain et al. [Jain et al., 2020] identified fire occurrence and susceptibility as major areas of machine learning research, while fewer studies have focused on predicting the eventual size of an already detected fire.

Second, many of the studies that predict fire size have been developed in regions such as Alaska, Siberia and the western United States [Coffield et al., 2019, Malashin et al., 2025, Pang et al., 2025]. Since wildfire behaviour is influenced by local climate, fuel and landscape characteristics, results from these regions cannot simply be assumed to apply to New Zealand.

Third, there is an opportunity to use the New Zealand Wildfire Occurrence Database for machine learning-based prediction of final burned area. Kiely et al. [Kiely et al., 2026] provide the data infrastructure for this type of research by combining wildfire records with weather, fuel, topographic and spatial information. However, the database itself does not answer the question of whether these variables can be used to predict the final size of a fire.

Fourth, the rarity of large fires creates an important modelling challenge. Research on imbalanced regression shows that standard regression methods may not give enough attention to rare target values [Camacho et al., 2022]. This is particularly important for wildfire applications because the largest fires can have much greater practical consequences than the many small fires.

Fifth, the way models are validated needs careful consideration. Studies such as Gholami et al. [Gholami et al., 2021] and Koh [Koh, 2023] show the importance of considering temporal and spatial dependence. This is particularly relevant to the WOD because both the number of observations and the availability of some predictor variables change over time.

Finally, historical wildfire records are not perfect. The WOD authors note that historical records can contain errors and uncertainty [Kiely et al., 2026]. Suppression actions are also not fully represented in the available information. Therefore, a small burned area does not necessarily mean that environmental

conditions were unsuitable for fire growth; it may also reflect early detection or effective suppression.

This research addresses these gaps by using the New Zealand WOD to investigate continuous prediction of final wildfire burned area. The study focuses on information available at or around the time of detection, compares different regression approaches, considers the strong imbalance in the target distribution and evaluates model performance separately for larger fires. A chronological validation strategy will also be used to provide a more realistic assessment of how the models perform on later wildfire records.

The main contribution of the research is not expected to be a completely new machine learning algorithm. Instead, the research aims to evaluate how existing machine learning approaches perform when applied to a New Zealand-specific wildfire dataset, with particular attention to the challenge of predicting rare large fires.

2.5 Chapter Summary

This chapter reviewed the background of wildfire prediction in New Zealand and the use of machine learning in wildfire research. Traditional approaches such as the Fire Weather Index provide useful information about environmental fire danger, but they do not directly predict the final burned area of an individual wildfire.

Previous studies have applied machine learning to wildfire occurrence, susceptibility, fire danger, fire spread and burned-area prediction. However, these studies differ considerably in their prediction targets, data sources, geographic regions and the information available when predictions are made. Research on final fire size is particularly relevant to this study, but much of the existing work has been carried out outside New Zealand.

The literature also shows that rare large fires create an important challenge for modelling. A standard regression model may perform well on the many small fires but struggle with the relatively small number of extreme cases. Recent research on imbalanced regression and extreme wildfire prediction provides several possible ways of addressing this issue.

Based on these findings, this research will use the New Zealand Wildfire Oc-

currence Database to investigate whether machine learning can predict the final burned area of a wildfire using information available at or around the time of detection. The next chapter describes the methodology used for this research, including data preprocessing, feature selection, target transformation, chronological validation, model selection and approaches for handling rare large-fire cases.

Chapter 3

Methodology

This chapter describes the dataset, exploratory data analysis, preprocessing steps, and machine learning methods that will be used to predict wildfire burned area in New Zealand. The main focus of the study is to predict the final burned area of a wildfire using information that is available at or before the time the fire is detected. This is intended to make the prediction useful for identifying fires that may develop into larger events.

3.1 Dataset

3.1.1 Dataset Description

This study uses the New Zealand Wildfire Occurrence Database (WOD), which contains information about wildfire events and their associated environmental, weather, topographical, spatial, and temporal conditions. The published WOD paper by Kiely et al. [Kiely et al., 2026] describes wildfire records from 1931 to 2022. However, the database version used in this study contains records from 1931 to 2025. Therefore, the analysis in this research is based on the available database through 2025 rather than being limited to the period reported in the original publication.

The dataset contains information about the location and timing of each wildfire, burned area, ignition source, fuel characteristics, terrain conditions, weather variables, and fire-weather indices. These variables provide information about

the environmental conditions associated with each recorded wildfire and can be used to investigate factors related to the final burned area.

In this study, *AreaHa*, representing the burned area in hectares, is used as the continuous target variable. The aim is to predict the final burned area of a wildfire using information available at or around the time the fire was detected. The remaining variables are considered as potential predictor variables after removing identifiers, target-related variables, and variables that may contain information generated after detection.

3.1.2 Exploratory Data Analysis

Exploratory data analysis (EDA) is carried out to understand the main characteristics of the wildfire data before developing the prediction models. The analysis focuses on the distribution of wildfire burned area, the availability of the predictor variables, relationships between environmental conditions and burned area, and spatial and temporal patterns in the data.

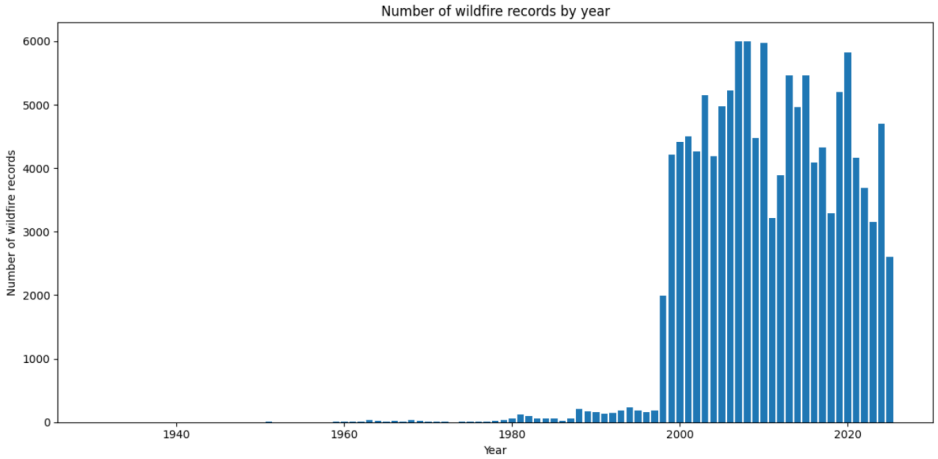


Figure 3.1: Number of wildfire records by year.

The number of wildfire records varies considerably across the study period. The database used in this study contains records from 1931 to 2025, but only a small number of records are available in the earlier years. The number of records gradually becomes larger from the 1980s and increases substantially from the late 1990s onwards, with several thousand records available in most recent years.

This uneven distribution of records is important when analysing temporal patterns and will be considered when selecting the training and testing periods.

The next part of the EDA examines the distribution of the target variable, *AreaHa*. Understanding this distribution is important because the objective of this study is to predict the final burned area of a wildfire as a continuous value.

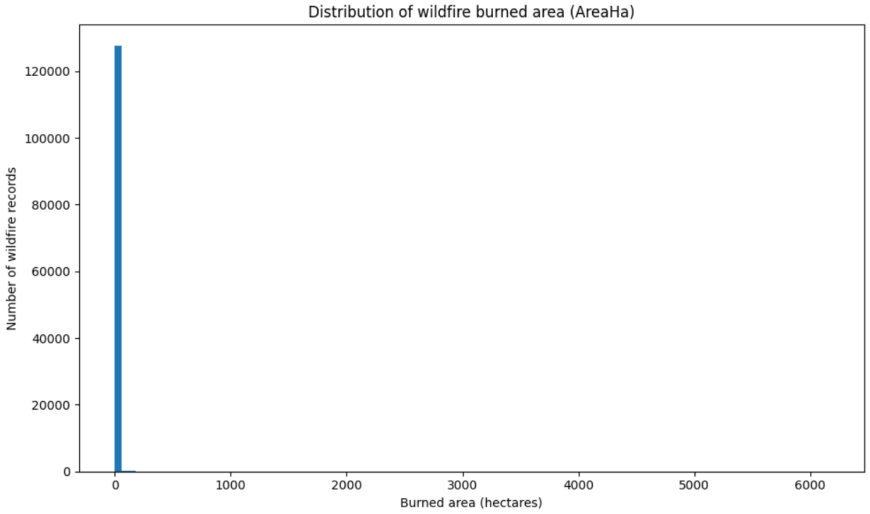


Figure 3.2: Distribution of wildfire burned area (*AreaHa*).

The distribution of *AreaHa* is highly right-skewed. Most wildfire records have a very small burned area, while only a small number of fires have very large burned areas. This creates a long tail in the target distribution, which is particularly important for this study because the larger fires are the main focus of the prediction problem.

Because of this strong skewness, a logarithmic transformation was also examined. The transformation $\log(AreaHa + 1)$ reduces the effect of extremely large values while still allowing observations with zero burned area to be included.

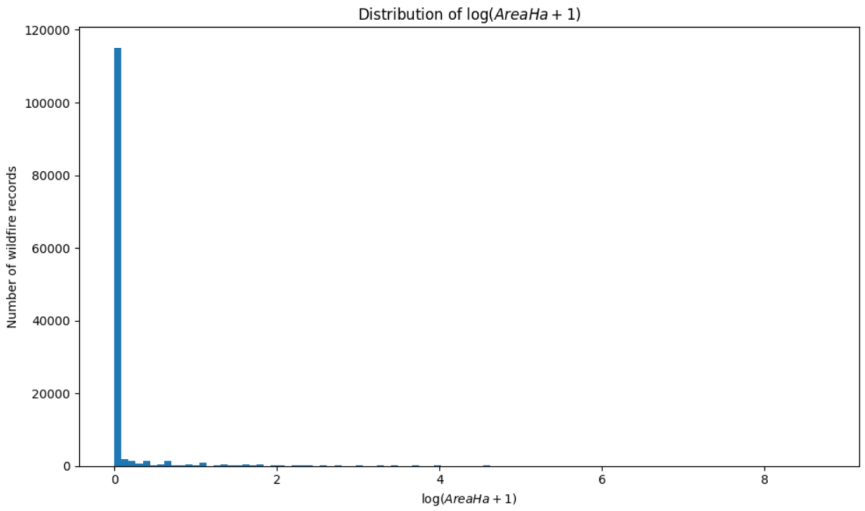


Figure 3.3: Distribution of $\log(AreaHa + 1)$.

The transformed distribution is easier to visualise, although it still contains a long right tail. Both the original and transformed target will therefore be considered during model development to determine which approach gives better predictions, especially for larger fires.

To understand the imbalance in fire sizes more clearly, the wildfire records were also grouped into different burned-area ranges.

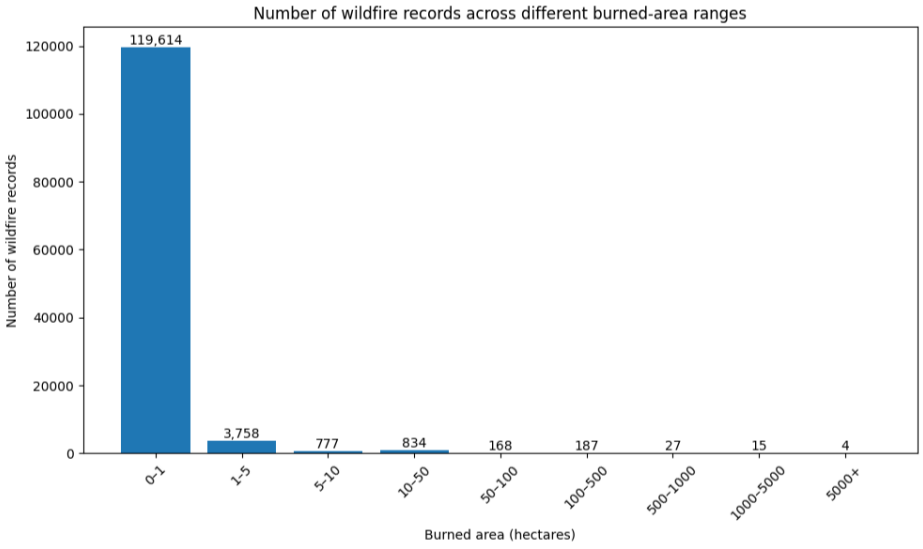


Figure 3.4: Number of wildfire records across different burned-area ranges.

The majority of wildfire records fall within the smallest burned-area range, while fires larger than 100 hectares are very rare. This shows that the regression target is highly uneven, even though the problem is not a classification problem. As a result, a model could achieve a low overall error by predicting small fires well while performing poorly on the relatively rare large fires. This will be considered when selecting the modelling and evaluation approaches.

The availability of the predictor variables was then examined to identify missing values and other data-quality issues.

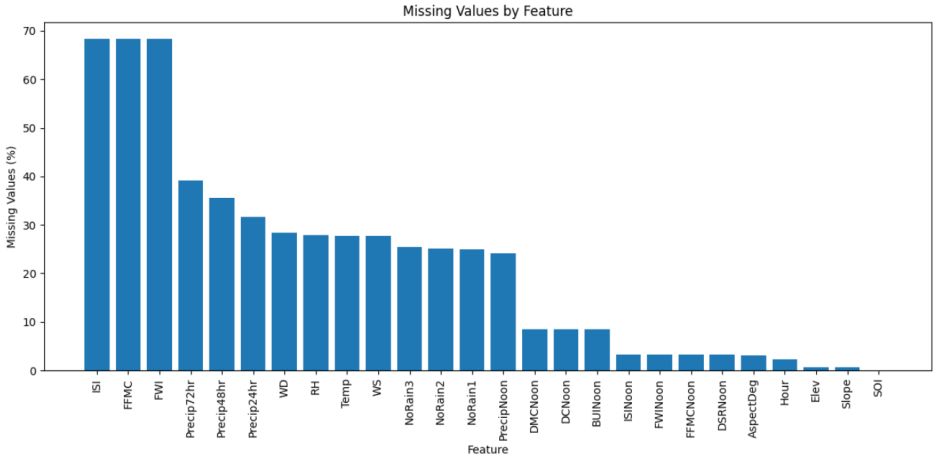


Figure 3.5: Percentage of missing values across the selected predictor variables.

Missing values are present in several weather and fire-weather variables. FWI, FFMC and ISI have particularly high levels of missingness, while variables such as elevation and slope have much fewer missing values. These differences need to be considered when deciding which variables to use and how missing values will be handled during preprocessing.

The availability of selected variables was also examined across different years, as the overall percentage of missing values does not show whether the missingness is concentrated in particular periods.

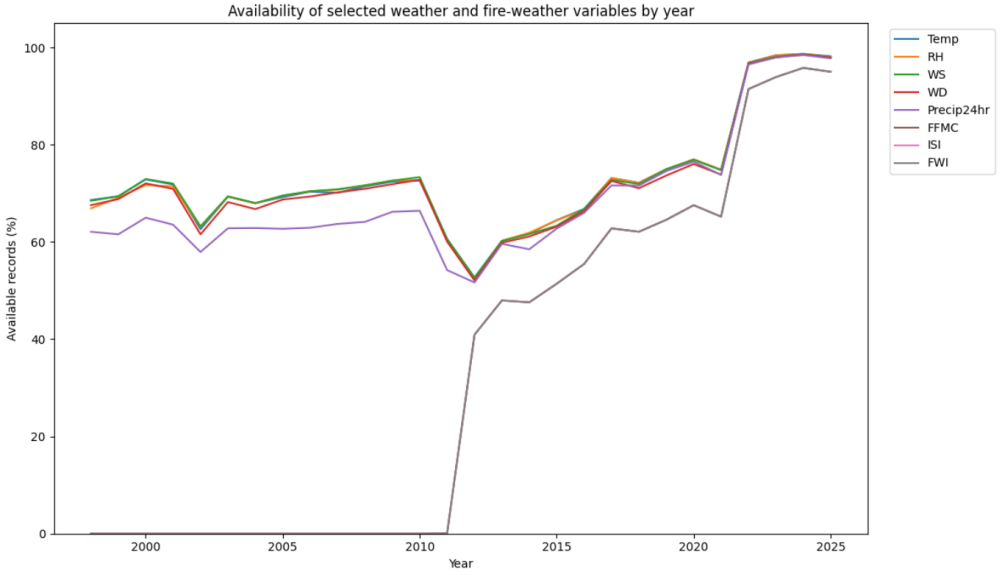


Figure 3.6: Availability of selected weather and fire-weather variables across the study period.

The availability of several variables changes considerably over time. In particular, FWI, FFMC and ISI have limited availability in the earlier years and become more consistently available in later years. This is important because the change in data availability may affect feature selection and model performance. It also supports the use of a chronological train-test split rather than randomly dividing the observations.

The relationships between burned area and selected environmental variables were then examined. Since the target variable is highly skewed, $\log(AreaHa+1)$ was used in the scatter plots to make the relationships easier to observe.

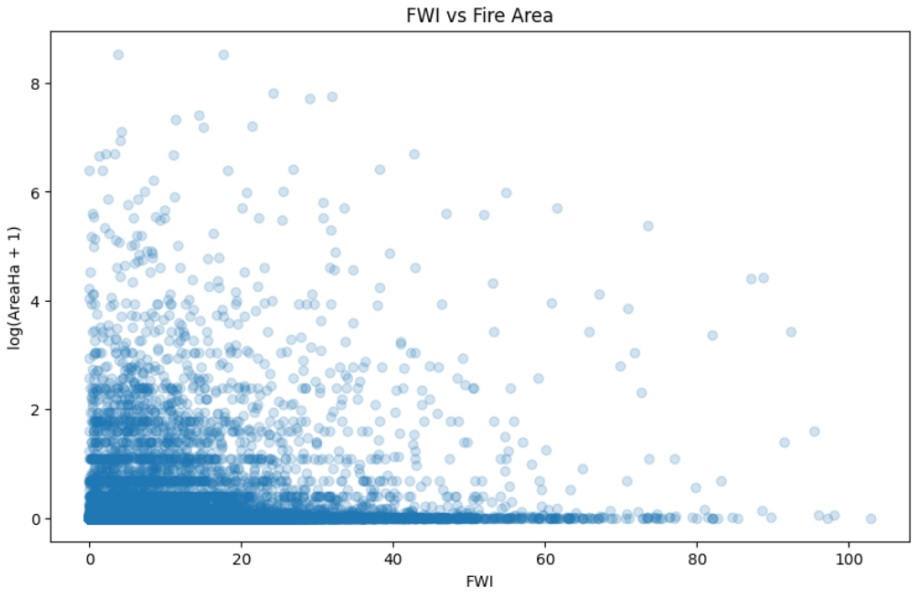


Figure 3.7: Relationship between Fire Weather Index (FWI) and $\log(AreaHa+1)$.

FWI shows a positive association with burned area, with some larger fires occurring at higher FWI values. However, the relationship is highly scattered, and large fires can also occur at lower FWI values. This suggests that FWI alone is not enough to predict the final burned area and that other environmental and spatial factors need to be considered together.

Relative humidity was also examined because it showed a negative association with burned area.

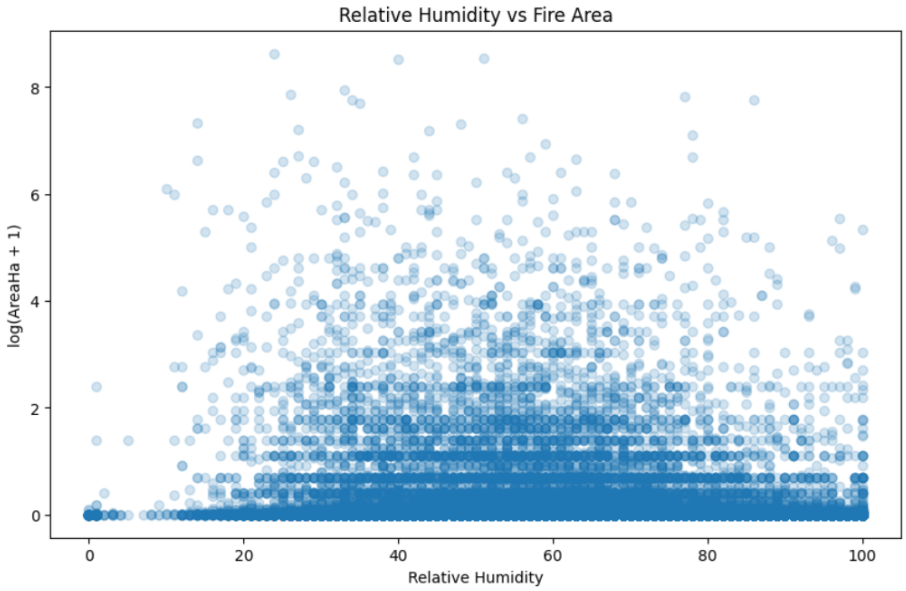


Figure 3.8: Relationship between relative humidity and $\log(AreaHa + 1)$.

The plot shows a weak negative relationship between relative humidity and burned area, with some larger fires occurring under lower humidity conditions. However, there is considerable variation in the observations, so relative humidity alone cannot explain the final size of a wildfire. Similar patterns were observed for other environmental variables such as temperature, wind speed, precipitation, slope and elevation, where no single variable showed a strong relationship with burned area. This supports the use of a multivariable machine learning approach.

Temporal patterns in wildfire size were also examined. Because the median burned area is close to zero for most years, the 90th and 99th percentiles were used to better represent changes in the upper part of the fire-size distribution.

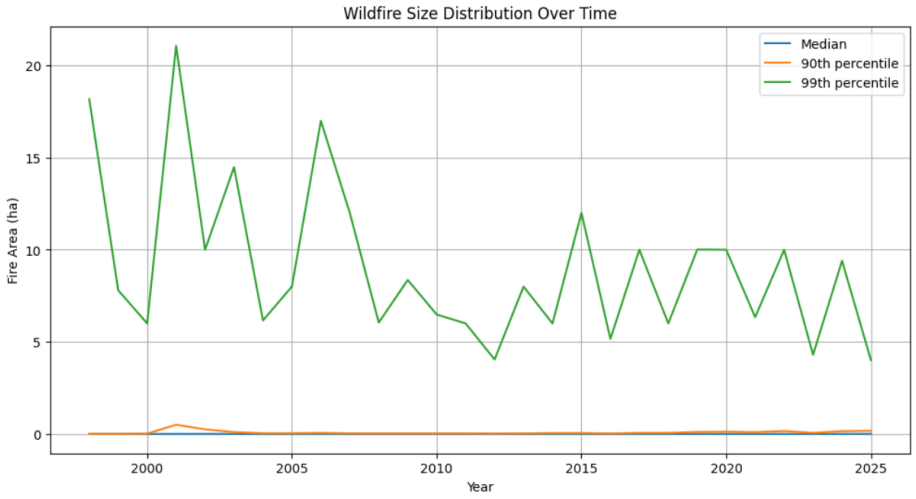


Figure 3.9: Upper-tail wildfire burned area by year using the 90th and 99th percentiles.

The 90th and 99th percentiles show variation in the upper tail of wildfire size across the years. However, the number of records is much smaller in the earlier years, so these differences should be interpreted carefully. The analysis is useful for understanding how the distribution of larger fires changes over time and further supports the need for a time-based evaluation strategy.

Categorical variables such as *FuelType*, *IgnSource*, and *FENZDist* will also be explored using frequency plots and burned-area comparisons. The fuel analysis shows that pasture grasslands and non-fuel areas make up a large proportion of the records. An unusual value such as "9999" in the fuel data will be investigated during preprocessing to determine whether it represents an unknown or missing category rather than a meaningful fuel type.

Finally, the spatial distribution of wildfire records was examined to identify possible geographical patterns in the dataset.

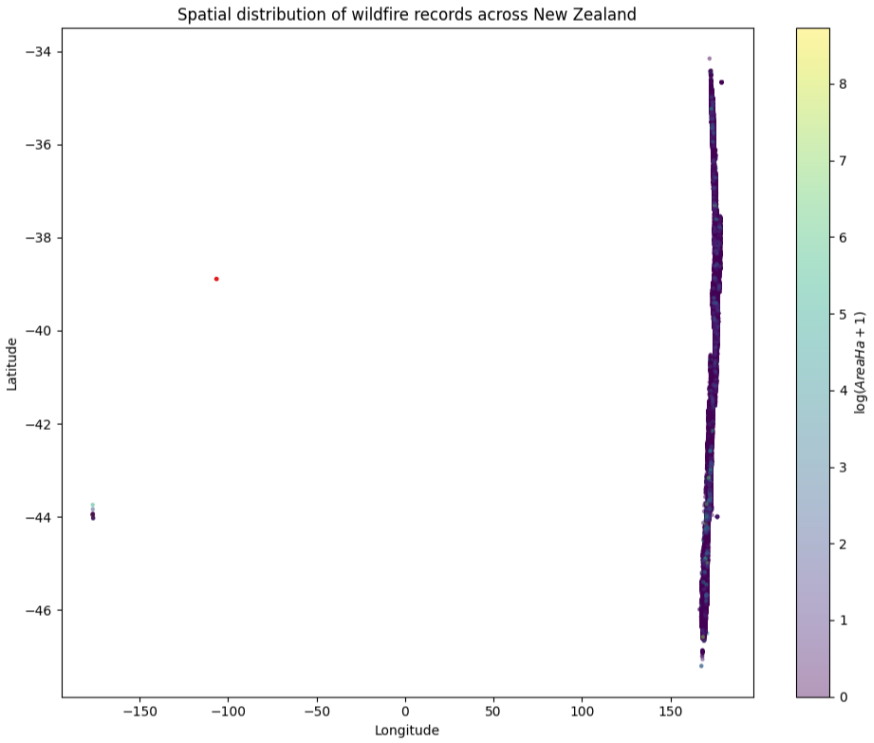


Figure 3.10: Spatial distribution of wildfire records across New Zealand, with burned area represented on a logarithmic scale.

The spatial distribution shows that wildfire records are spread across different parts of New Zealand, with most observations located on the mainland. This provides an initial indication that location may contain useful information for the prediction task. Latitude, longitude and other spatial variables will therefore be considered as potential predictors.

Overall, the EDA shows that the main challenges in the dataset are the highly skewed burned-area distribution, the small number of large fires, missing values in several predictor variables, and changes in data availability across time. These findings will guide the data preprocessing, feature selection, data splitting and modelling decisions described in the following sections.

3.2 Data Preprocessing

The preprocessing steps will be based on the findings from the exploratory data analysis. The aim is to prepare the dataset while retaining as much useful information as possible and avoiding the introduction of information that would not be available at the time of wildfire detection.

3.2.1 Data Cleaning

First, records containing invalid or placeholder values will be identified. In particular, values of *-9999* will be treated as missing rather than as actual measurements. Other suspicious values and categories will also be checked before modelling.

Duplicate records will be identified and removed where appropriate. Variables that identify the record rather than describe the wildfire environment, such as database IDs and organisation identifiers, will not be used as predictive features.

The target variable, *AreaHa*, will also be checked for missing or invalid values. Records without a usable target value cannot be used for supervised regression and will therefore be removed from the modelling dataset.

3.2.2 Handling Missing Values

Missing values will be handled according to the type and amount of missingness in each variable. Numerical variables with a moderate amount of missing data will be considered for imputation using statistics calculated from the training data, such as the median. Median imputation is preferred for highly skewed environmental variables because it is less affected by extreme values than mean imputation.

Categorical variables will be considered for mode imputation or an explicit "Unknown" category where appropriate.

Variables with very high levels of missingness will not automatically be imputed. Instead, their usefulness and missing-data pattern will be examined first. This is particularly important for variables such as FWI, FFMC and ISI, whose availability changes substantially across the historical period.

All imputation parameters will be estimated using the training data only. This will prevent information from the validation or test periods from leaking into the training process.

3.2.3 Feature Selection

The initial feature set will focus on information that could realistically be available at or before wildfire detection.

The main candidate weather variables are temperature, relative humidity, wind speed, wind direction, precipitation and recent rainfall indicators. Fuel-related variables such as fuel type, fuel class and land cover will also be considered. Topographic variables include elevation, slope, aspect and landform. Spatial and temporal variables such as latitude, longitude, month, hour, day and season will also be evaluated.

Fire-weather indices such as FFMC, ISI, FWI, DMC, DC, BUI and DSR will be investigated separately. Although these variables may contain useful information about fire-weather conditions, some of them have substantial missingness in the historical dataset. In addition, variables that represent fire behaviour after detection will be excluded if they are found to contain information that would not have been available at prediction time.

Variables such as HFI and ROS will therefore be treated with particular caution because they may contain information related to fire behaviour itself rather than only the environmental conditions available when the fire is detected.

The final feature set will be selected based on data availability, relevance to the research question, potential information leakage, and model performance.

3.2.4 Target Transformation

Because *AreaHa* has a highly skewed distribution, two approaches will be considered.

The first approach will predict the original burned area directly:

$$AreaHa = f(X)$$

The second approach will predict the logarithmically transformed target:

$$\log(AreaHa + 1) = f(X)$$

For the second approach, predictions will be converted back to hectares using the inverse transformation. The two approaches will be compared using the same test data to determine which provides more useful predictions, particularly for larger fires.

3.2.5 Data Splitting

Since the dataset contains wildfire records collected over many years, a random train-test split will not be used as the main evaluation strategy. A random split could allow information from similar periods to appear in both training and testing data and may give an overly optimistic estimate of performance.

Instead, a chronological split will be used. Earlier years will be used for model development, and later years will be kept for testing. The exact cutoff will be selected after confirming the final version and time range of the dataset.

A validation set or time-based validation strategy will be used during model development for model selection and hyperparameter tuning. The final test period will remain untouched until the models and preprocessing decisions have been finalised.

3.2.6 Handling the Imbalanced Regression Problem

Although this study uses regression rather than classification, the distribution of the target is still highly imbalanced. Most observations correspond to very small fires, while large fires are rare.

This creates a risk that a model may obtain a good overall error by predicting small fire areas well while performing poorly on the larger fires. Therefore, model evaluation will not rely on a single overall error measure.

Methods for imbalanced regression will be investigated during the modelling stage. In particular, resampling approaches designed for continuous targets, such as Geometric SMOTE for regression and related methods for rare target values, will be considered. These methods have been proposed to improve the prediction of rare and extreme values in continuous regression prob-

lems [Camacho et al., 2022]. Recent wildfire research has also applied imbalanced regression and resampling methods to the prediction of final burned area at the time of ignition [Camacho et al., 2026].

Resampling will only be applied to the training data. The validation and test datasets will retain the original distribution of fire sizes so that model performance can be evaluated under realistic conditions.

3.3 Machine Learning Model Selection

Several regression algorithms will be considered in this study. The models will be selected to provide a comparison between simpler interpretable methods and more flexible non-linear machine learning approaches.

3.3.1 Baseline Model

A simple baseline regression model will first be developed. A baseline is important because it provides a reference point for determining whether more complex machine learning models provide a meaningful improvement.

A simple model such as Linear Regression will be used as the initial baseline. The baseline will be trained using the same training and validation procedure as the other models.

3.3.2 Tree-Based Machine Learning Models

Tree-based ensemble methods will be considered because the relationship between wildfire size and environmental conditions is unlikely to be purely linear. Variables such as temperature, humidity, wind, fuel type and terrain may interact with each other.

Random Forest Regression will be considered as one of the initial non-linear models. It can model non-linear relationships and interactions without requiring strong assumptions about the relationship between the input variables and burned area.

Gradient boosting models such as XGBoost or LightGBM will also be considered. These models are suitable for structured tabular datasets and can capture

complex relationships between multiple environmental variables.

The final set of models will be selected based on the results obtained during experimentation rather than assuming that one algorithm will perform best in advance.

3.3.3 Model Evaluation

Model performance will be evaluated using several regression metrics. Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) will be used to measure overall prediction error.

However, because the target distribution is highly skewed, overall performance alone will not be sufficient. Additional analysis will examine model performance for larger fires and for different ranges of burned area. This will help determine whether a model that performs well for the majority of small fires is also capable of identifying and estimating larger fires.

Where appropriate, metrics will also be calculated on the transformed target and after converting predictions back to hectares.

The final model will be selected based on both overall predictive performance and its ability to predict the upper tail of the wildfire-size distribution.

3.3.4 Model Interpretation

Model interpretation will also be considered important because the purpose of the study is not only to produce predictions but also to understand which environmental conditions are associated with larger wildfire outcomes.

Feature importance and, where appropriate, SHAP-based explanations will be used to investigate the contribution of important features to model predictions. This will help determine whether variables such as weather, fuel, terrain or temporal conditions are consistently important for predicting wildfire burned area.

3.4 Chapter Summary

This chapter has described the proposed methodology for predicting wildfire burned area in New Zealand using machine learning. The study will use the New Zealand Wildfire Occurrence Database and treat burned area in hectares as a continuous regression target.

The exploratory analysis has shown that the target is highly right-skewed, with most fires being very small and only a small number of fires developing into large events. The analysis has also identified substantial missingness in some fire-weather variables and changes in feature availability over time.

Based on these findings, the preprocessing stage will focus on data cleaning, missing-value handling, feature selection, target transformation and a time-based data splitting strategy. Different regression models will then be compared, with particular attention given to the prediction of rare large fires rather than relying only on overall average performance.

The following chapter will present the experimental results and compare the performance of the selected machine learning models.

Chapter 4

Experimental Design and Expected Results

This chapter describes how the proposed machine learning models will be tested and compared. Since the aim of this research is to predict wildfire burned area as a continuous value, the experiments will focus on regression performance. Particular attention will be given to the prediction of large fires because they are rare in the dataset and are more important for the main objective of this study.

4.1 Experimental Setup

The experiments will be carried out using Python. Libraries such as Pandas and NumPy will be used for data processing, while Scikit-learn will be used for preprocessing, baseline models and evaluation. Tree-based models such as Random Forest and gradient boosting models will also be considered during the experiments.

The same preprocessing pipeline will be applied consistently across the models. Missing values will be handled using information from the training data only, and categorical variables will be encoded where required. Any resampling method used to address the imbalanced target distribution will also be applied only to the training data. This is important to prevent information from the test data from being introduced into the training process.

The dataset will be divided chronologically rather than randomly. Earlier

observations will be used for model training and validation, while later observations will be kept as the final test set. This better represents the real situation where a model is trained using historical wildfire records and then used to predict the size of future fires. The exact year used as the boundary will be selected after the final feature set and usable time period have been confirmed.

During model development, the training data will be further divided or evaluated using a time-based validation strategy. The validation data will be used for model selection and hyperparameter tuning, while the final test period will not be used until the models and preprocessing decisions have been finalised.

Several experiments will be carried out to compare different modelling approaches. The first experiment will use the original burned-area target. The second will use $\log(AreaHa + 1)$ as the target. The effect of imbalanced-regression methods will then be investigated by comparing models trained with and without resampling or other approaches designed for rare target values.

The main experiments will therefore compare:

- A simple baseline model;
- Standard regression models without target balancing;
- Regression models using the logarithmically transformed target;
- Regression models using methods for handling rare and large target values;
- The performance of these approaches on both the overall dataset and the large-fire subset.

This comparison will help determine whether improving the prediction of rare large fires comes at the cost of performance on the majority of small fires.

4.2 Evaluation Metrics

Since the target variable is continuous, regression metrics will be used to evaluate the models. More than one metric will be reported because no single metric can fully describe the performance of a model on a highly skewed target such as wildfire burned area.

4.2.1 Mean Absolute Error (MAE)

Mean Absolute Error (MAE) measures the average absolute difference between the actual and predicted burned area. It is easy to interpret because the error is expressed in the same unit as the target variable.

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{4.1}$$

where y_i is the actual burned area and $\hat{y}_i$ is the predicted burned area.

MAE will be used as one of the main evaluation metrics because it gives a direct measure of how far the predictions are from the actual burned areas.

4.2.2 Root Mean Squared Error (RMSE)

Root Mean Squared Error (RMSE) measures the square root of the average squared prediction error.

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{4.2}$$

RMSE gives more weight to large errors than MAE. This is useful in this study because a very large prediction error for a large wildfire can be more important than a small error for one of the many very small fires.

4.2.3 R-Squared (R^2)

R^2 will also be reported to provide an overall measure of how much of the variation in burned area is explained by the model.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{4.3}$$

where $\bar{y}$ is the mean of the actual burned area.

Although R^2 provides a useful overall measure, it will not be used as the only metric because the wildfire burned-area distribution is highly skewed and is dominated by very small fires.

4.2.4 Performance on Larger Fires

Overall metrics may hide poor performance on large fires because large fires make up only a very small part of the dataset. Therefore, model performance will also be evaluated separately for different fire-size ranges.

For example, MAE and RMSE will be calculated for groups such as fires below 1 hectare, fires between 1 and 100 hectares, and larger fires above 100 hectares. The exact groups may be adjusted depending on the final test-set distribution.

This will allow the study to answer an important question: whether a model that performs well overall is also able to predict the relatively rare large fires.

Performance on the upper tail of the target distribution will be given particular attention because predicting large wildfire outcomes is the main motivation for this research.

4.2.5 Evaluation of Log-Transformed Models

For models trained using $\log(AreaHa + 1)$, predictions will be converted back to the original hectare scale before calculating the main MAE and RMSE values. This allows the results of models trained using the original and transformed targets to be compared in the same units.

Additional error measures may also be calculated on the transformed target to understand how well the model performs across the full distribution.

4.3 Validation Strategy

The validation strategy is important because the wildfire records are collected over time. A random cross-validation approach could place observations from similar time periods in both the training and validation sets and may give an overly optimistic estimate of how the model will perform on future wildfire records.

4.3.1 Chronological Train-Test Split

A chronological split will therefore be used as the main evaluation strategy. Earlier years will be used for training and model development, while later years will be held out as the final test period.

The exact split will be determined after the final preprocessing and feature selection steps. This is important because some predictor variables have much lower availability in the earlier part of the dataset. The selected split should provide enough training data while also giving a meaningful period for testing the model on later wildfire records.

The test set will not be used during model development. It will only be used after the final model and preprocessing approach have been selected.

4.3.2 Time-Based Validation

Time-based validation will be used within the training period for model selection and hyperparameter tuning. In each validation step, the model will be trained using earlier observations and validated using observations from a later period.

For example, an earlier group of years can be used for training and the following years for validation, with the validation period moving forward over subsequent iterations. This keeps the temporal order of the data and better represents the way the model would be used in practice.

4.3.3 Hyperparameter Tuning

Hyperparameters will be tuned using only the training and validation data. Grid search or random search may be used depending on the number of parameters and computational cost.

The final test set will remain untouched during this process. Once the best model and hyperparameters have been selected, the model will be evaluated once on the final test period.

4.3.4 Validation of Imbalanced Regression Methods

The effect of handling the imbalanced target distribution will also be tested. A standard model will first be trained without resampling. This result will then be compared with models using an approach designed for continuous and rare target values.

Geometric SMOTE for regression and related imbalanced-regression methods will be considered because these methods are designed for situations where rare values in a continuous target are important [Camacho et al., 2022]. This is also relevant to the wildfire problem, as recent research has applied imbalanced-regression techniques to predicting final wildfire burned area [Camacho et al., 2026].

Any resampling will be performed only on the training data. The validation and test sets will keep their original distribution so that the evaluation remains representative of the actual wildfire data.

4.4 Expected Results

The expected results are based on the findings from the EDA and the previous research discussed in the literature review. Since the burned-area target is highly skewed and most fires are very small, a simple regression model is expected to perform reasonably on small fires but may have difficulty predicting the rare large fires.

The Linear Regression baseline is expected to provide a useful reference point but may not capture the non-linear relationships between wildfire size and environmental conditions. Tree-based models such as Random Forest and gradient boosting models are expected to perform better because they can capture non-linear relationships and interactions between weather, fuel, terrain, spatial and temporal variables.

The logarithmic transformation of the target is also expected to help the models deal with the strong right-skew in burned area. However, this does not necessarily mean that the log-transformed model will produce the best predictions for the largest fires. Therefore, its performance will be compared with models trained directly on the original burned-area values.

It is also expected that methods designed for imbalanced regression may improve predictions for rare large fires. However, improving performance on large fires may result in slightly higher errors for the much larger number of small fires. This trade-off will be examined rather than choosing a model only on the basis of its overall MAE or RMSE.

The study does not assume that a particular model will achieve a specific level of accuracy before the experiments are completed. Instead, the aim is to identify which modelling approach provides the best balance between overall prediction accuracy and performance on larger wildfire events.

The results may provide useful information about whether environmental and spatial conditions available around the time of wildfire detection can be used to estimate the eventual burned area. If the models are able to identify conditions associated with larger fire outcomes, the results could support further development of early wildfire risk assessment tools in New Zealand.

There are also several limitations that may affect the results. The final burned area recorded in the database may contain some measurement or reporting errors because fire size can be estimated after an event. In addition, the database does not contain detailed information about suppression actions, which may affect how large a fire eventually becomes. A fire may remain small because it was quickly detected and suppressed even when environmental conditions were favourable for fire growth.

Another limitation is the changing availability of some predictor variables over time. Variables such as FWI, FFMC and ISI have considerably less coverage in some earlier years. This may limit the amount of historical data that can be used when these variables are included. The temporal distribution of the wildfire records may also affect how well the model generalises to future years.

These limitations will be considered when interpreting the final results rather than treating the model predictions as direct evidence of causation.

4.5 Chapter Summary

This chapter described how the proposed regression models will be tested and compared. The experiments will use a chronological evaluation strategy so that the final test period represents future wildfire records rather than randomly se-

lected observations from the same period as the training data.

MAE, RMSE and R^2 will be used to measure overall regression performance, while additional evaluation will focus on larger fires because they are rare but important to the research objective. Different target transformations and approaches for handling the imbalanced burned-area distribution will also be compared.

The expected results are that non-linear tree-based models will capture relationships that are difficult for a simple linear baseline to model. However, the main focus will be on finding whether the proposed methods can improve prediction of large fires without losing too much performance on the majority of small fires. The results will also be interpreted in the context of missing data, fire-size measurement, suppression effects and changes in data availability over time.

Chapter 5

Conclusion

Wildfires in Aotearoa New Zealand can vary greatly in their final burned area, making it difficult to estimate the potential size of a fire soon after it is detected. This research proposes the use of machine learning to predict the final burned area of an individual wildfire using information available at or around the time of detection. Instead of treating fire size as a set of categories, the study will treat burned area in hectares as a continuous prediction problem. The New Zealand Wildfire Occurrence Database (WOD) will be used because it brings together wildfire records with weather, fuel, topographic, spatial and temporal information [Kiely et al., 2026].

The research will first examine the WOD to understand the distribution of burned area, missing values and changes in feature availability over time. Several regression models will then be developed and compared using a chronological train-test split. This approach is intended to provide a more realistic evaluation of how the models may perform when predicting later wildfire records. Since most of the fires in the dataset are very small and only a small number become large, the study will also investigate the problem of imbalanced regression. Approaches such as target transformation and resampling methods will be considered, with particular attention given to whether they improve predictions for the rare large fires [Camacho et al., 2022, Camacho et al., 2026].

There are several challenges that may affect the results. One of the main issues is the large amount of missing data in some fire-weather variables. The availability of these variables also changes considerably over the years, which

means that using every available feature could make the model less useful for earlier records or introduce differences between training and testing data. The study will therefore consider feature availability when selecting predictors and will perform imputation using information from the training data only. Variables that are not available at the prediction time will also be excluded to reduce the risk of information leakage.

Another limitation is the quality of the burned-area measurements. Historical wildfire records may contain errors or uncertainty, and the final recorded area may not always perfectly represent the actual fire size [Kiely et al., 2026]. The dataset also does not fully describe firefighting and suppression activities. This is important because a small burned area may result from effective and rapid suppression rather than from environmental conditions that were unsuitable for fire growth. These factors cannot be completely removed using the available data, so they will be considered when interpreting the model results.

The strong imbalance in fire size is another important limitation. A model may perform well on the large number of small fires while still producing poor predictions for the few large fires. For this reason, the research will not rely only on overall error measures. Performance on different burned-area ranges will also be examined to determine whether the models are able to provide useful predictions for larger events. Feature importance and SHAP analysis will further be used to understand which environmental and geographical variables contribute to the model predictions.

Overall, this research aims to determine how well machine learning can predict the final burned area of wildfires in New Zealand using information available at or around detection. The expected outcome is not necessarily a single model that works for every wildfire, but a clearer understanding of which modelling approaches are suitable for this problem, how the rare large-fire cases can be handled, and what limitations need to be considered when using the WOD for prediction. The findings could provide a useful starting point for future research into data-driven wildfire decision support in Aotearoa New Zealand.

Bibliography

Ardid, A., Power, A., Valencia, A., Pearce, H. G., Gross, S., Dempsey, D., and Boer, M. (2026). From forecast skill to economic value: sub-hourly wildfire potential forecasting across australian regions. *International Journal of Wildland Fire*, 35(4).

Ardid, A., Valencia, A., Power, A., Boer, M. M., Katurji, M., Gross, S., and Dempsey, D. (2025). Sub-hourly forecasting of fire potential using machine learning on time series of surface weather variables. *International Journal of Wildland Fire*, 34(1).

Breiman, L. (2001). Random forests. *Machine Learning*, 45:5–32.

Camacho, L., Catry, F. X., and Bacao, F. (2026). An integrated resampling and machine learning framework for predictive analytics of large wildfires. *Decision Analytics Journal*, 18:100674.

Camacho, L., Douzas, G., and Bacao, F. (2022). Geometric smote for regression. *Expert Systems with Applications*, 193:116387.

Chen, T. and Guestrin, C. (2016). Xgboost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794. ACM.

Cisneros, D., Gong, Y., Yadav, R., Hazra, A., and Huser, R. (2023). A combined statistical and machine learning approach for spatial prediction of extreme wildfire frequencies and sizes. *Extremes*, 26:301–330.

Coffield, S. R., Graff, C. A., Chen, Y., and Smyth, P. (2019). Machine learning to

predict final fire size at the time of ignition. *International Journal of Wildland Fire*, 28(11):861–873.

D'Arcy, E., Murphy-Barltrop, C. J. R., Shooter, R., and Simpson, E. S. (2023). A marginal modelling approach for predicting wildfire extremes across the contiguous united states. *Extremes*, 26:381–398.

Di Giuseppe, F., McNorton, J., Lombardi, A., and Wetterhall, F. (2025). Global data-driven prediction of fire activity. *Nature Communications*, 16:2918.

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *The Annals of Statistics*, 29(5):1189–1232.

Gholami, S., Kodandapani, N., Wang, J., and Lavista Ferres, J. (2021). Where there's smoke, there's fire: Wildfire risk predictive modeling via historical climate data. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 35, pages 15309–15315.

Jaafari, A., Zenner, E. K., Panahi, M., and Shahabi, H. (2019). Hybrid artificial intelligence models based on a neuro-fuzzy system and metaheuristic optimization algorithms for spatial prediction of wildfire probability. *Agricultural and Forest Meteorology*, 266–267:198–207.

Jain, P., Coogan, S. C. P., Subramanian, S. G., Crowley, M., Taylor, S., and Flannigan, M. D. (2020). A review of machine learning applications in wildfire science and management. *Environmental Reviews*, 28(4):478–505.

Kiely, L., Gross, S., Woods, D., Christensen, B., Guo, F., and Lee, E. (2026). A spatial database of wildfire occurrence and associated fire environment conditions in new zealand. *Scientific Data*.

Koh, J. (2023). Gradient boosting with extreme-value theory for wildfire prediction. *Extremes*, 26:273–299.

Kondylatos, S., Prapas, I., Ronco, M., Papoutsis, I., Camps-Valls, G., Piles, M., and Carvalhais, N. (2022). Wildfire danger prediction and understanding with deep learning. *Geophysical Research Letters*, 49(17):e2022GL099368.

Lampe, S., Gudmundsson, L., Kraft, B., Hantson, S., Kelley, D., Humphrey, V., Le Saux, B., Chuvieco, E., and Thiery, W. (2026). Burnn (v1.0): a data-driven fire model. *Geoscientific Model Development*, 19(2):955–988.

Lundberg, S. M. and Lee, S.-I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems*, volume 30.

Malashin, I. P., Masich, I., Nelyub, V., Borodulin, A., Gantimurov, A., and Tynchenko, V. (2025). Assessing wildfire extents in siberian forests using machine learning. *Scientific Reports*, 15:32834.

Melia, N., Pearce, H. G., Salinger, M. J., et al. (2022). Aotearoa new zealand's 21st-century wildfire climate. *Earth's Future*.

Ministry for the Environment (2025). Wildfire risk in new zealand, 1997–2023. Technical Report CR 615, Ministry for the Environment, Wellington, New Zealand.

Pang, B., Cheng, S., Huang, Y., Jin, Y., Guo, Y., Prentice, I. C., Harrison, S. P., and Arcucci, R. (2025). Fire-image-densenet (fidn) for predicting wildfire burnt area using remote sensing data. *Computers & Geosciences*, 195:105783.

Pearce, H. G., Salinger, M. J., and Renwick, J. A. (2007). Impact of climate variability on fire danger. Technical Report 72, New Zealand Fire Service Commission, Wellington, New Zealand.

Renwick, J. A., Salinger, M. J., Zheng, X., and Pearce, H. G. (2007). Prediction of fire weather and fire danger. Technical Report 83, New Zealand Fire Service Commission, Wellington, New Zealand.

Shmuel, A. and Heifetz, E. (2022). Global wildfire susceptibility mapping based on machine learning models. *Forests*, 13(7):1050.

Tonini, M., D'Andrea, M., Biondi, G., Degli Esposti, S., Trucchia, A., and Fiorucci, P. (2020). A machine learning-based approach for wildfire susceptibility mapping. the case study of the liguria region in italy. *Geosciences*, 10(3):105.